



Adama Science and Technology University

School of Electrical Engineering and Computing

Department of Computer Science and Engineering

Internship Report

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Duration of Internship: 2.5 Months

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Executive Summary

This report summarizes my internship at AFRICOM PLC from June 10 to September 22, 2025, where I worked as a front-end developer on a multi-tenant Human Resource Management System (HRMS). The internship bridged academic learning with professional practice by providing hands-on experience in building scalable and secure enterprise software.

My role focused on designing and implementing responsive UI components, developing dynamic interfaces, and integrating front-end features using React, Next.js, Tailwind CSS, and React Hook Form with Zod. I collaborated closely with back-end developers to ensure smooth API integration and consistent user experiences across multiple tenants.

Key challenges included applying modern front-end architecture and managing complex state. Through research and problem-solving, I gained practical insights into component-based design, form validation, and collaborative workflows.

Overall, the internship enhanced my technical and professional skills while reinforcing my career path in front-end development. To improve future programs, I recommend providing structured onboarding, advanced knowledge-sharing sessions, and greater exposure to UI/UX design processes.

Acknowledgement

I would like to express my deepest gratitude to Adama Science and Technology University (ASTU), particularly the School of Electrical Engineering and Computing, for providing me with the opportunity to undertake this internship program. The internship is an essential part of our academic journey, and I am truly grateful for the platform it created to connect classroom learning with real-world practice.

I am especially thankful to my academic supervisor, INS Shummet Fekadesilase, for his constant support, encouragement, and valuable feedback throughout the entire internship period. His guidance not only helped me remain focused on my objectives but also inspired me to approach challenges with confidence and persistence.

My heartfelt appreciation also goes to AFRICOM PLC for welcoming me into their organization and offering a highly professional environment where I could gain hands-on experience as a back-end developer. I am particularly indebted to my industry supervisor, CTO Eskindir Teklu, whose mentorship was instrumental in helping me navigate technical tasks, understand enterprise-level software development, and grow both technically and professionally.

I also extend my sincere thanks to my colleagues and team members at AFRICOM PLC, who were always willing to collaborate, share their expertise, and provide encouragement throughout the project. Working alongside them on the Human Resource Management System (HRMS) with Multi-Tenancy Support project allowed me to enhance my technical skills while also learning the value of teamwork and knowledge sharing.

Finally, I owe profound gratitude to my family and friends for their unwavering support, motivation, and encouragement during this journey. Their belief in me has been a source of strength and inspiration throughout the internship.

List of Acronyms

- **ASTU** : Adama Science and Technology University
- **IT** : Information Technology
- **ICT** : Information and Communication Technology
- **ERP** : Enterprise Resource Planning
- **HRMS** : Human Resource Management System
- **RBAC** : Role-Based Access Control
- **API** : Application Programming Interface
- **DBMS** : Database Management System
- **SQL** : Structured Query Language
- **JWT** : JSON Web Token
- **MVP** : Minimum Viable Product
- **BPO** : Business Process Outsourcing
- **ITO** : Information Technology Outsourcing
- **IoT** : Internet of Things
- **JS** : Javascript
- **CI/CD** : Continuous Integration / Continuous Deployment
- **KPI** : Key Performance Indicator

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Chapter 1: Introduction

The internship program bridges academic theory and industry practice, equipping students with real-world skills under professional mentorship. At Adama Science and Technology University (ASTU), it is a core requirement for Electrical Engineering and Computing students, designed to prepare them for professional careers.

As part of this program, we worked on the design and development of a multi-tenant Human Resource Management System (HRMS). The system provides secure, scalable, and customizable solutions for managing employees, attendance, payroll, performance, and recruitment. My primary role was front-end development, collaborating with another front-end developer to ensure responsive, user-friendly interfaces integrated seamlessly with back-end services built in Node.js.

This documentation outlines the objectives, scope, expected outputs, and organizational background, while reflecting on lessons learned, challenges faced, and professional growth.

1.1 General Overview

Objective: The internship aimed to provide hands-on experience in enterprise-level software development, focusing on front-end engineering. Key responsibilities included:

- Building responsive and user-friendly interfaces.
- Integrating front-end components with Node.js APIs.
- Applying UI/UX and front-end architecture best practices.
- Collaborating with teammates under real project conditions.

Scope: Between June 10 – September 22, my tasks covered:

- Designing UIs for HRMS modules (employee management, attendance, leave, recruitment, dashboards).
- Ensuring responsiveness and accessibility.
- Collaborating with back-end developers for API integration.

- Debugging, testing, and documenting front-end components.
- Contributing to planning, feedback, and improvement sessions.

Expected Outputs: Interns at AFRICOM are expected to:

- Apply academic knowledge in practical IT projects.
- Deliver quality work on time.
- Document work accurately and respond to feedback.
- Contribute to AFRICOM's mission of delivering top-tier IT solutions.

1.2 Background of the Organization

Description: Established in 2004 G.C., AFRICOM Technologies is one of Ethiopia's pioneering ISO-certified IT service providers. With nearly two decades of experience, it serves both government and private sectors, recognized for professionalism, technical expertise, and innovative solutions. Its vision is to become a continental leader in IT while driving digital transformation in Ethiopia and Africa.

Mission: AFRICOM is committed to delivering innovative, secure, and high-quality IT solutions that enhance organizational efficiency and competitiveness. It seeks to bridge business needs with technology, foster digital transformation, and build long-term partnerships through professionalism, integrity, and continuous improvement.

Products and Services: AFRICOM offers a wide range of IT services, including:

- MIS consultancy for business process optimization.
- Enterprise software development (front-end & back-end).
- BPO services to streamline operations.
- Single Window Projects for integrated service delivery.
- IT auditing for reliability and compliance.
- E-commerce solutions to support online businesses.

Chapter 2: Management, training, and employee information

2.1 Major Duty

During my internship at AFRICOM Technologies, I contributed as a front-end developer to a multi-tenant Human Resource Management System (HRMS), building responsive interfaces for employee management, payroll, attendance, performance, and recruitment. The internship included training in HTML, CSS, JavaScript, and React, hands-on development with React and Tailwind CSS, collaboration via Git, and integration with back-end services built on ASP.NET Core 9 and Next-Auth. The final phase focused on testing, debugging, and refining the system, enhancing both my technical and teamwork skills in full-stack development.

2.2 Orientation of Responsibilities

The team emphasized professionalism, accountability, and timely delivery, while my supervisor, CTO Eskindir Teklu, provided guidance, resources, and feedback to align my work with project goals. I contributed to API development, database management, and secure coding, gaining a comprehensive understanding of enterprise software development, teamwork, and skills in time management, organization, and communication.

2.3 Supervisor Ensures

CTO Eskindir Teklu offered consistent mentorship through technical advice, tool recommendations, and detailed code reviews. His guidance improved my coding, project management, and communication skills while encouraging continuous learning.

2.4 Qualities Developed

During my internship, I developed **technical skills** in React, Next.js, Entity Framework Core, and Next-Auth, while improving **problem-solving**, **project management**, and **secure coding** through RBAC and multi-tenancy implementation. I strengthened **time management**, **team collaboration**, and **professionalism**, consistently delivering quality work and meeting deadlines.

Chapter 3: Specific Job Information

3.1 Technical Duties

At AFRICOM Technologies, my primary role was front-end development with back-end integration for the HRMS. Key tasks included:

- Building responsive UIs with React, Next.js, and Tailwind CSS.
- Handling forms using React Hook Form with Zod for validation.
- Integrating APIs for seamless data flow.
- Implementing authentication and RBAC with Next-Auth.
- Collaborating with back-end developers to ensure consistency.
- Managing state, optimizing performance, and debugging.
- Using Git for version control and team collaboration.
- Applying best practices in modern front-end development.

3.2 Other Responsibilities

Beyond technical work, I:

- Collaborated with team members and shared feedback.
- Documented design decisions, code, and challenges.
- Followed company coding and workflow standards.
- Managed time effectively to meet deadlines.
- Engaged in continuous learning and maintained professionalism.

3.3 Technical Knowledge Gained

Skills developed during the internship included:

- Form Handling & Validation: React Hook Form + Zod.
- API Integration: Smooth front-end/back-end communication.
- Authentication & RBAC: Secure access with Next-Auth.

- State Management: Efficient component interaction.
- Version Control: Git for collaboration and quality assurance.
- Testing & Debugging: Improving usability and reliability.
- Documentation: Supporting teamwork and project continuity.

3.4 Relevant Projects

I contributed to building a multi-tenant HRMS with three portals:

- Super Admin: Organization setup and monitoring.
- Admin User: Employee management, attendance, recruitment, performance.
- Employee: Dashboards, leave requests, training, and announcements.

My role focused on:

- Designing user-friendly, responsive interfaces.
- Integrating APIs (Node.js and .NET 9) for real-time updates.
- Developing dashboards and HR workflow forms.
- Implementing RBAC for secure access.
- Participating in agile sprints, testing, and documentation.

Impact: The HRMS delivered a scalable and secure solution for multi-organization HR operations. My contributions enhanced usability, performance, and accessibility, while strengthening my expertise in front-end engineering, API integration, and enterprise software development.

Chapter 4: Reflection and Conclusion

4.1 Career Goal During Internship

This internship supported my goal of becoming a front-end developer, providing hands-on experience in responsive UI design, form handling, and API integration with React, Next.js, React Hook Form, and Zod. I also practiced version control, debugging, and teamwork, building a solid foundation for creating scalable, maintainable interfaces.

4.2 Change of Career Goals

Initially, my focus was only on front-end development. However, I realized its deep connection with back-end systems, APIs, and authentication. This broadened my goal to not only coding but also understanding system integration, performance optimization, and secure access control, aiming to contribute strategically to application design.

4.3 Value of Internship

The internship bridged theory and practice, enhancing my confidence in front-end development while improving my skills in workflows, teamwork, and problem-solving. It also reinforced my commitment to pursuing a career in this field.

4.4 Challenges Faced

Key challenges included adapting to new frameworks, mastering form handling with API integration, meeting tight deadlines, and debugging complex issues. These improved my time management, problem-solving, and resilience under pressure.

4.5 Evaluation and Critique

I gained strengths in problem-solving, UI/UX detail, collaboration, punctuality, and technical skills such as component design, validation, and state management. To improve, I aim to master advanced frameworks, performance optimization, and large-scale applications through continuous learning and practice, working toward becoming a versatile front-end developer.

Chapter 5: Recommendations

5.1 Recommendation about the Organization

AFRICOM is an innovative ICT company that provides a professional yet supportive environment. The team was approachable and offered mentorship whenever I faced challenges. The internship gave me valuable exposure to real-world ICT operations, especially in front-end development with Next.js, React, React Hook Form + Zod, and Next-Auth. Tasks like building UIs, validating forms, and integrating APIs enriched my skills far beyond theory. Compared to many programs, this internship offered exceptional opportunities for growth and industry-standard experience, strengthening both my technical and soft skills.

5.2 Recommendation about the Internship Program

The internship was highly valuable, but a few improvements could enhance learning:

- Provide structured onboarding with tutorials on frameworks, design systems, and UI/UX best practices.
- Organize regular workshops on emerging front-end technologies, performance optimization, and accessibility.
- Increase mentorship and feedback sessions for better task alignment and guidance.
- Offer better access to resources such as internal docs, design assets, and sandbox projects.
- Review stipend and benefits to support interns financially and boost focus.

Implementing these changes would help interns gain stronger skills and be more industry-ready.

5.3 Suggestion

I strongly recommend AFRICOM for future ASTU student placements. Its innovative culture, mentorship, and meaningful projects provide interns with hands-on experience in modern front-end practices, problem-solving, and teamwork. The company ensures a balance between support and professional responsibility, making it an excellent choice for students seeking practical exposure in ICT.

Annexes

Annex A: Design / Frontend Pages

A)Login Page : This page allows users to authenticate into the HRMS system. It validates user credentials and sends a request to the back-end API for JWT authentication.

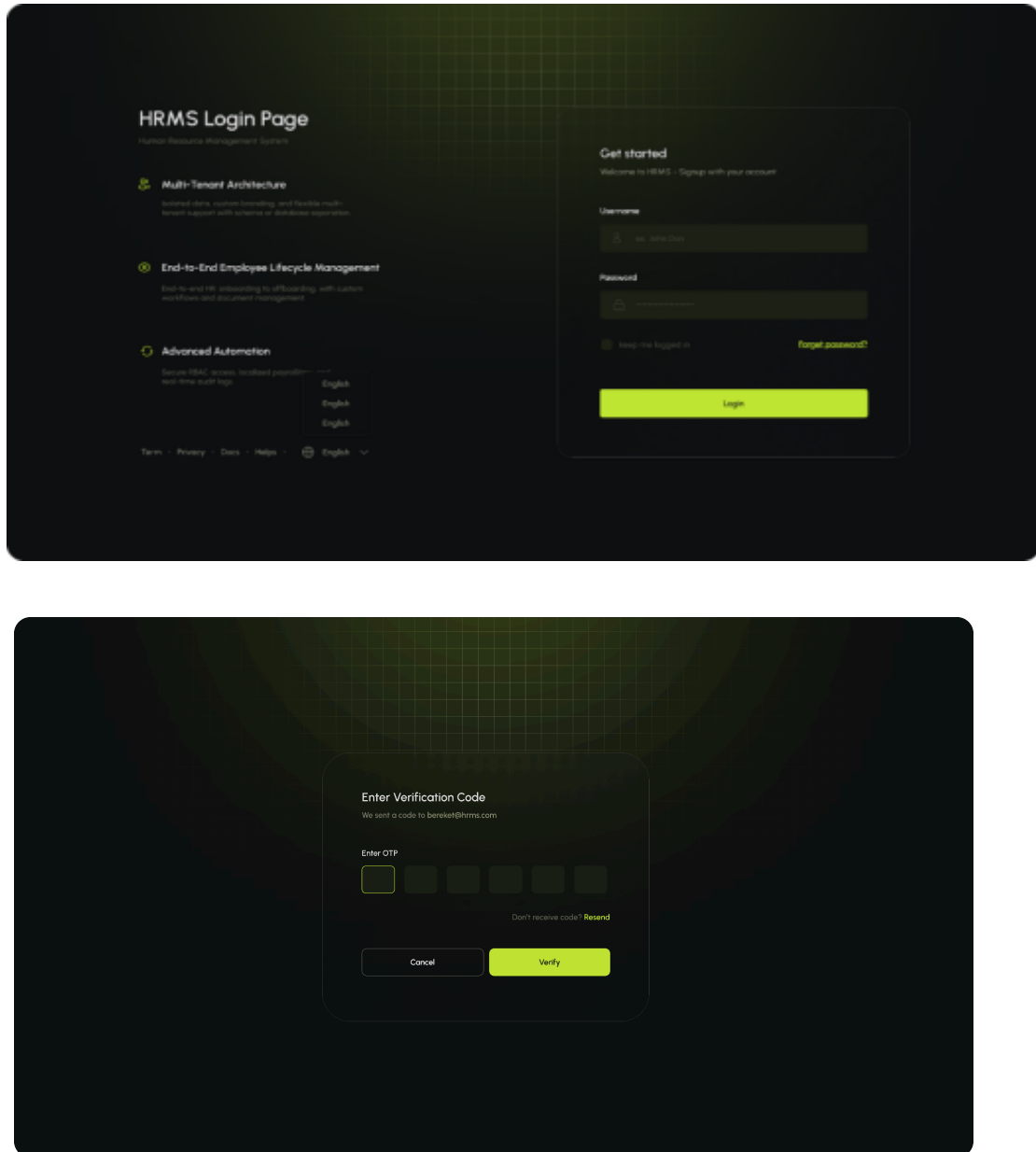


Fig.1 Login-page

B) Super Admin portal: Manage all organizations, configure organization settings, create new organizations, monitor usage statistics, and handle super admin tasks.

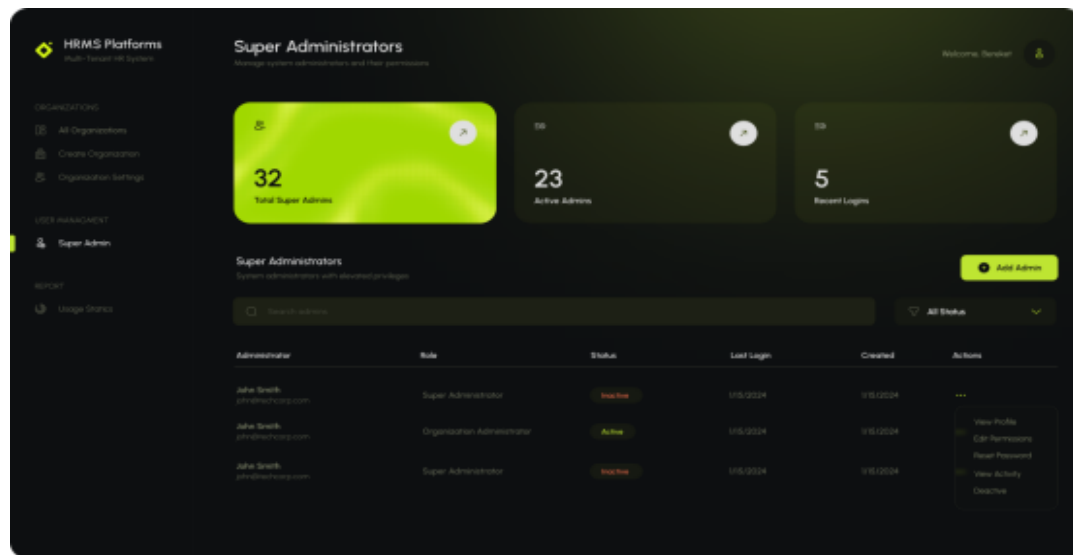


Fig.2 The Super-admin portal

C) Admin User portal:

Core: Dashboard, organization management, employee management, attendance tracking, and leave management.

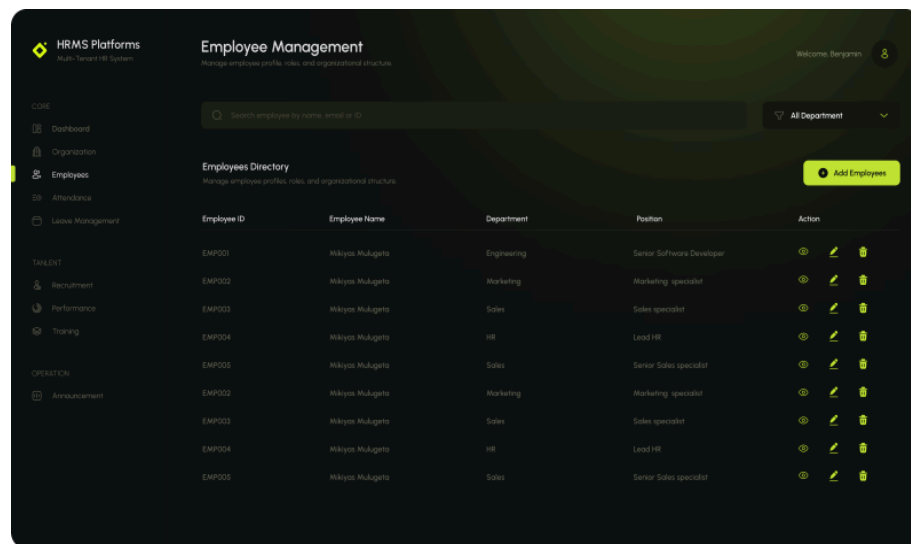


Fig.3 The Employee page.

Fig.4 The Employee page to add new employee

Talent: Recruitment, performance evaluation, and training modules.

Operation: Announcement management.

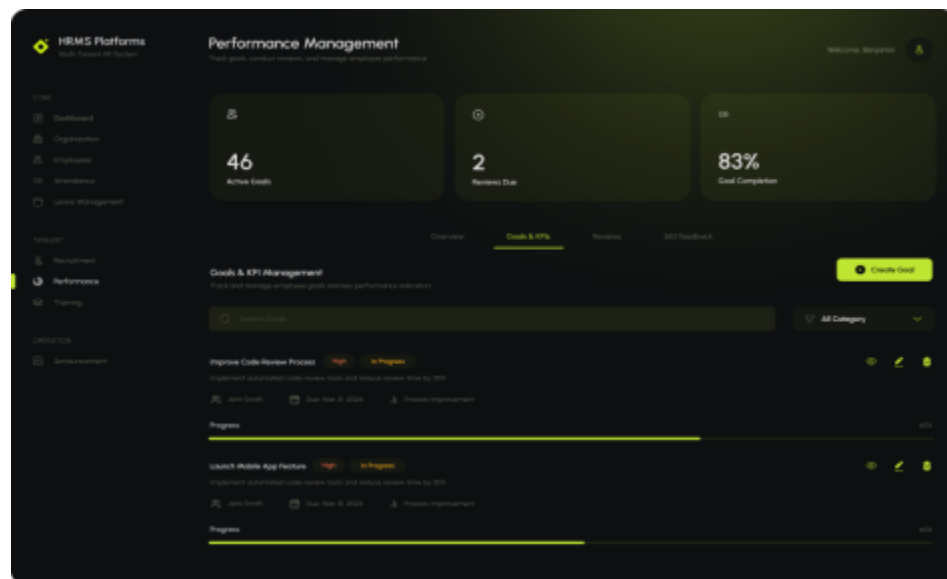


Fig.5 The Performance page to evaluate performance of employees.

D) Employee Portal:

Core: Dashboard, attendance records, and leave requests.

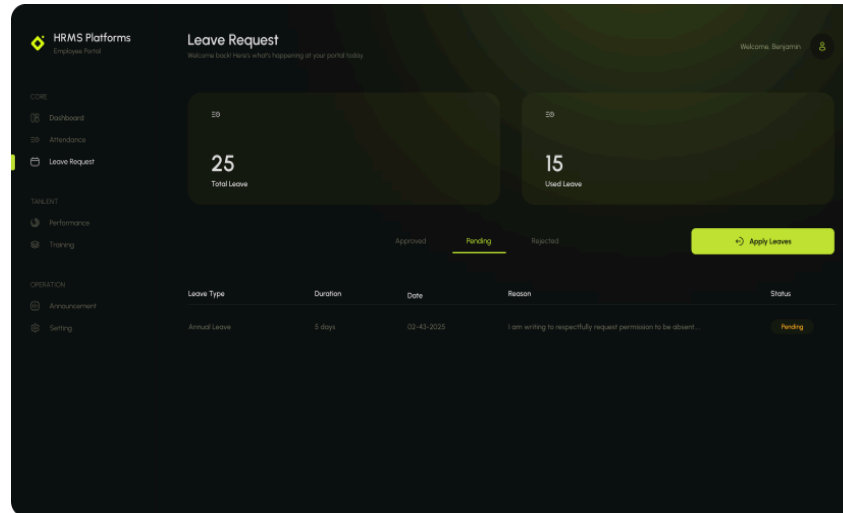


Fig.6 The Leave Request page in Employee Portal page to ask to leave.

Talent: Performance review and training participation.

Operation: Announcements and user settings.

Annex B: Database Diagram: The diagram illustrates the structure of the HRMS database, showing the tables for **Employees**, **Tenants**, **Roles**, **Leaves**.

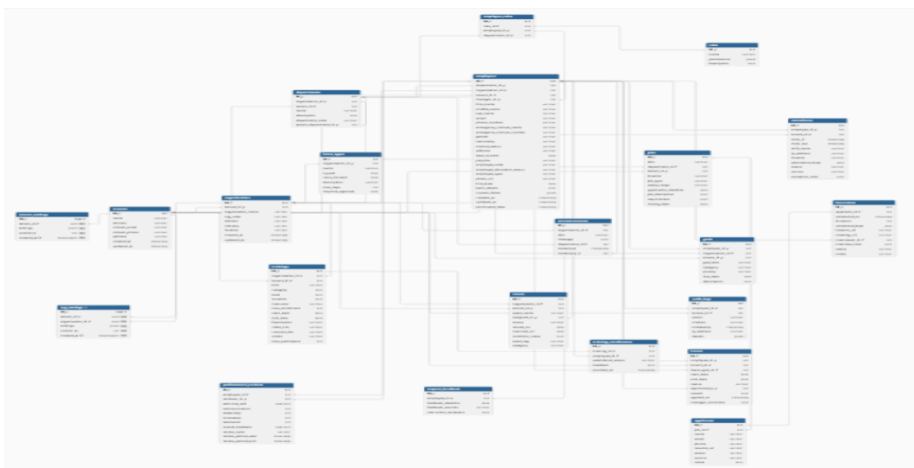


Fig.7 DB Diagram

Annex C: Postman API Testing

POST Employee Details : Example of a POST request to Create an employee

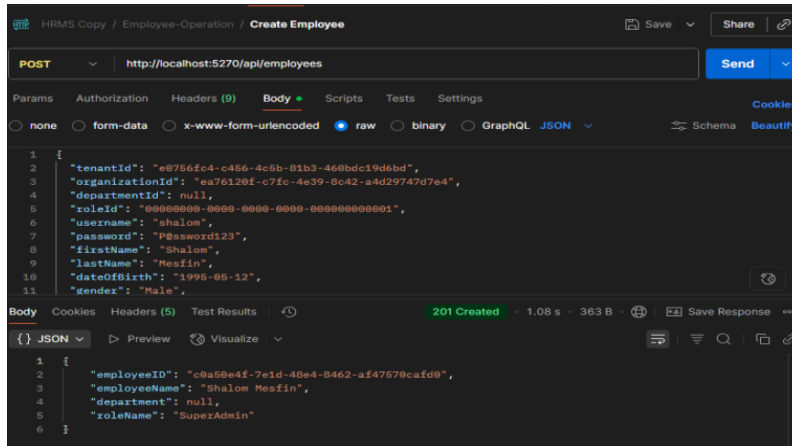


Fig.8 Postman API Testing

Annex D: Sample Code from Model

Description: This code snippet represents the **Employee model** in the backend, defining properties, relationships, and navigation properties for the database.

```
using System;

using System.ComponentModel.DataAnnotations;
using System.ComponentModel.DataAnnotations.Schema;

namespace HRMS.Backend.Models
{
    [Table("employees")]
    public class Employee
```

```

{
    [Key]
    [Column("id")]
    public Guid EmployeeID { get; set; } = Guid.NewGuid();
    // FKs
    [Column("department_id")]
    public Guid? DepartmentId { get; set; } // ← now nullable
    [Required]
    [Column("organization_id")]
    public Guid OrganizationId { get; set; }
    [Required]
    [Column("tenant_id")]
    public Guid TenantId { get; set; }
    [Required]
    [Column("role_id")]
    public Guid RoleId { get; set; }
    // Auth / access
    [Required, MaxLength(100)]
    [Column("username")]
    public string Username { get; set; } = string.Empty; // ← required
    [Required]
    [Column("password_hash")]
    public string PasswordHash { get; set; } = string.Empty; // ← required (store a
hash, not plain text)
    // Personal info
    [Required, MaxLength(100)]
    [Column("first_name")]
    public string FirstName { get; set; } = string.Empty;
    [Required, MaxLength(100)]
    [Column("last_name")]

```

```

public string LastName { get; set; } = string.Empty;
[Required, MaxLength(50)]
[Column("gender")]
public string Gender { get; set; } = string.Empty;
[Required, MaxLength(100)]
[Column("nationality")]
public string Nationality { get; set; } = string.Empty;
[Required, MaxLength(50)]
[Column("marital_status")]
public string MaritalStatus { get; set; } = string.Empty;
[Required, MaxLength(500)]
[Column("address")]
public string Address { get; set; } = string.Empty;
[Required]
[Column("date_of_birth")]
public DateTime DateOfBirth { get; set; }
// Contact
[Required, MaxLength(255)]
[Column("email")]
public string Email { get; set; } = string.Empty;
[Required, MaxLength(50)]
[Column("phone_number")]
public string PhoneNumber { get; set; } = string.Empty;
[Required, MaxLength(200)]
[Column("emergency_contact_name")]
public string EmergencyContactName { get; set; } = string.Empty;
[Required, MaxLength(50)]
[Column("emergency_contact_number")]
public string EmergencyContactNumber { get; set; } = string.Empty;
// Job

```

```

[Required, MaxLength(150)]
[Column("job_title")]
public string JobTitle { get; set; } = string.Empty;
[Required, MaxLength(100)]
[Column("employee_education_status")]
public string EmployeeEducationStatus { get; set; } = string.Empty;
[Required, MaxLength(50)]
[Column("employee_type")]
public string EmploymentType { get; set; } = string.Empty;
[Required, MaxLength(2083)]
[Column("photo_url")]
public string PhotoUrl { get; set; } = string.Empty;
[Required]
[Column("hire_date")]
public DateTime HireDate { get; set; }
[MaxLength(50)]
[Column("employee_code")]
public string? EmployeeCode { get; set; } // required via controller rule but will be
auto-generated if missing

```

```

// Payroll / misc

```

```

[Required]
[Column("bank_details")]
public string BankDetails { get; set; } = "{}";
[Required]
[Column("custom_fields")]
public string CustomFields { get; set; } = "{}";
[Column("benefits_enrollment")]
public string? BenefitsEnrollment { get; set; } // ← now nullable

```

```

        [Column("shift_details")]
        public string? ShiftDetails { get; set; }    // ← now nullable
        // Timestamps
        [Required]
        [Column("created_at")]
        public DateTime CreatedAt { get; set; } = DateTime.UtcNow;
        [Required]
        [Column("updated_at")]
        public DateTime UpdatedAt { get; set; } = DateTime.UtcNow;
        [Column("terminated_date")]
        public DateTime? TerminatedDate { get; set; }
        [Column("leave_credit")]
        public decimal LeaveCredit { get; set; } = 20; // default 20 days
        public DateTime LastCreditUpdate { get; set; } = DateTime.UtcNow;
        // Navs
        public Tenant Tenant { get; set; } = null!;
        public Organization Organization { get; set; } = null!;
        public Department? Department { get; set; } // optional now
        public Role Role { get; set; } = null!;

        public ICollection<Attendance> Attendances { get; set; } = new
        List<Attendance>();
    }
}

```

Annex E: Sample Code from Controller

Description: This controller snippet demonstrates how the back-end handles **HTTP requests**, interacts with the database, and returns responses to the frontend using DTOs.

```

// POST: api/employees
[HttpPost]
public async Task<IActionResult> Create([FromBody] EmployeeCreateDto dto)
{
    if (!ModelState.IsValid) return ValidationProblem(ModelState);

    // Basic FK existence checks
    var tenantExists = await _context.Tenants.AnyAsync(t => t.Id == dto.TenantId);
    if (!tenantExists) return BadRequest($"Tenant {dto.TenantId} not found.");

    var org = await _context.Organizations.AsNoTracking()
        .FirstOrDefaultAsync(o => o.Id == dto.OrganizationId && o.TenantId
            == dto.TenantId);
    if (org is null) return BadRequest("Organization not found in the specified tenant.");

    if (dto.DepartmentId.HasValue)
    {
        var deptOk = await _context.Departments.AnyAsync(d =>
            d.Id == dto.DepartmentId.Value &&
            d.OrganizationId == dto.OrganizationId &&
            d.TenantId == dto.TenantId);
        if (!deptOk) return BadRequest("Department not found in the specified
            organization/tenant.");
    }
}

```



```

var roleExists = await _context.Roles.AnyAsync(r => r.Id == dto.RoleId);
if (!roleExists) return BadRequest($"Role {dto.RoleId} not found.");

// Uniqueness checks (per tenant)
var emailClash = await _context.Employees.AnyAsync(e => e.TenantId ==
dto.TenantId && e.Email == dto.Email);
if (emailClash) return Conflict(new { message = "Email already exists in tenant." });

// Username unique per tenant
var username = dto.Username.Trim();
var usernameClash = await _context.Employees.AnyAsync(e => e.TenantId ==
dto.TenantId && e.Username == username);
if (usernameClash) return Conflict(new { message = "Username already exists in
tenant." });

// EmployeeCode: generate if null/empty; ensure uniqueness per tenant
var employeeCode = string.IsNullOrEmpty(dto.EmployeeCode)
    ? await GenerateUniqueEmployeeCodeAsync(dto.TenantId, dto.FirstName,
dto.LastName)
    : dto.EmployeeCode!.Trim().ToUpperInvariant();

if (!string.IsNullOrEmpty(dto.EmployeeCode))
{

```

```
var codeClash = await _context.Employees.AnyAsync(e => e.TenantId ==  
dto.TenantId && e.EmployeeCode == employeeCode);  
if (codeClash) return Conflict(new { message = "Employee code already exists in  
tenant." });  
}
```

```
var entity = new Employee  
{  
    EmployeeID = Guid.NewGuid(),  
    TenantId = dto.TenantId,  
    OrganizationId = dto.OrganizationId,  
    DepartmentId = dto.DepartmentId,    // nullable OK  
    RoleId = dto.RoleId,
```

```
    FirstName = dto.FirstName.Trim(),  
    LastName = dto.LastName.Trim(),  
    DateOfBirth = dto.DateOfBirth,  
    Gender = dto.Gender.Trim(),  
    Nationality = dto.Nationality.Trim(),  
    MaritalStatus = dto.MaritalStatus.Trim(),
```

```
    Email = dto.Email.Trim(),  
    PhoneNumber = dto.PhoneNumber.Trim(),  
    Address = dto.Address.Trim(),  
    EmergencyContactName = dto.EmergencyContactName.Trim(),  
    EmergencyContactNumber = dto.EmergencyContactNumber.Trim(),
```

```

        JobTitle = dto.JobTitle.Trim(),
        EmploymentType = dto.EmploymentType.Trim(),
        EmployeeEducationStatus = dto.EmployeeEducationStatus.Trim(),
        PhotoUrl = dto.PhotoUrl.Trim(),
        // If JoiningDate is null, use current UTC date as default
        HireDate = dto.HireDate == default ? DateTime.UtcNow : dto.HireDate,

        EmployeeCode = employeeCode,
        Username = username,
        PasswordHash = HashPassword(dto.Password),

        BankDetails = string.IsNullOrEmpty(dto.BankDetails) ? "{}" :
dto.BankDetails,
        CustomFields = string.IsNullOrEmpty(dto.CustomFields) ? "{}" :
dto.CustomFields,
        BenefitsEnrollment = dto.BenefitsEnrollment,
        ShiftDetails = dto.ShiftDetails,
        LeaveCredit = dto.LeaveCredit,

        CreatedAt = DateTime.UtcNow,
        UpdatedAt = DateTime.UtcNow
    };

    _context.Employees.Add(entity);

```

```

await _context.SaveChangesAsync();

// Fetch the created employee along with department and role details
var createdEmployee = await _context.Employees
    .Where(e => e.EmployeeID == entity.EmployeeID)
    .Include(e => e.Department) // Include Department if it exists
    .Include(e => e.Role)      // Include Role
    .AsNoTracking()
    .Select(e => new
    {
        e.EmployeeID,
        EmployeeName = $"{e.FirstName} {e.LastName}",
        Department = e.Department != null ? e.Department.DepartmentName : null, //
Null if no department
        RoleName = e.Role.Name
    })
    .FirstOrDefaultAsync();

return CreatedAtAction(nameof(GetById), new { id = entity.EmployeeID },
createdEmployee);
}

```

Reference

- Internship Guidelines for interns
- Old practical attachments
- our supervisor
- <https://africom.et/>